

Romeo Mtsweni

Software Engineer | Computer Systems Engineer | AI & Embedded Systems Enthusiast

📍 Midrand, Gauteng, South Africa

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PROFESSIONAL PROFILE

Motivated and adaptable Software Engineer with a National Diploma in Intelligent Industrial Systems and currently completing an Advanced Diploma in Computer Systems Engineering at Tshwane University of Technology.

I enjoy building software that solves real-world problems through web technologies, artificial intelligence, embedded systems, networking, IoT, and industrial automation. I have practical experience developing enterprise applications, supporting automation platforms, configuring network infrastructure, and working within collaborative engineering teams.

I am passionate about continuous learning and excited to contribute to innovative software engineering teams while growing my technical expertise.

TECHNICAL SKILLS

Languages

Python • C# • Java • JavaScript • HTML5 • CSS3 • SQL • C++ •

Software Development

Object-Oriented Programming

REST APIs

Responsive Web Design

Git & GitHub

Software Testing

Debugging

Agile Development

Frameworks & Technologies

Bootstrap

Firebase

Unity

Arduino

ESP32

MATLAB

Proteus

Visual Studio

Visual Studio Code

Engineering

Embedded Systems

Internet of Things (IoT)

Industrial Automation

PLC Programming

Networking

Technical Support

Machine Learning

Artificial Intelligence

PROFESSIONAL EXPERIENCE

New Island Technologies

Solutions Engineer Intern

March 2025 – March 2026

Responsibilities

- Supported enterprise automation solutions.
- Assisted with Control-M implementations.
- Worked with enterprise scheduling platforms.
- Assisted customers with technical troubleshooting.
- Collaborated with solution engineers on enterprise software deployments.
- Gained experience in enterprise software environments and technical consulting.

ABT Telecoms

Software Developer & Network Technician Intern

August 2023 – June 2024

Responsibilities

- Assisted in development of the Zarfin enterprise web application.
 - Installed and configured routers across healthcare facilities.
 - Worked on a government connectivity project.
 - Installed solar backup systems for networking equipment.
 - Performed software support and troubleshooting.
 - Worked closely with software developers and networking teams.
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EDUCATION

Advanced Diploma

Computer Systems Engineering (2025 - Expected 2026)

Tshwane University of Technology

Relevant Modules

Artificial Intelligence

Machine Learning

Embedded Systems

Engineering Project Design

Digital Signal Processing

Research Methods

Computer Engineering

National Diploma

Intelligent Industrial Systems

Tshwane University of Technology (Completed 2024)

N4 Electrical Engineering

Tshwane North TVET College (Completed 2016)

National Senior Certificate

Hoërskool FH Odendaal (Completed 2014)

PROJECTS

Smart IoT Helmet

Designed and developed a smart safety helmet capable of monitoring environmental conditions and worker health using ESP32, Firebase, Wi-Fi and multiple sensors.

Technologies

ESP32 • Firebase • IoT • C++ • Embedded Systems

Zarfin Enterprise System

Contributed to the frontend development and maintenance of the Zarfin enterprise management platform.

Technologies

HTML

CSS

JavaScript

Bootstrap

Solar Tracking System

Developed an automatic solar tracking solution that improves solar panel efficiency through real-time light tracking.

Elevator Control System

Designed a digital elevator controller demonstrating industrial automation and control system concepts.

Unity Game Development

Developed a game using Unity, Python and C# during academic coursework.

CORE COMPETENCIES

- ✓ Software Engineering
 - ✓ Artificial Intelligence
 - ✓ Machine Learning
 - ✓ Embedded Systems
 - ✓ Internet of Things
 - ✓ Enterprise Software
 - ✓ Networking
 - ✓ Industrial Automation
 - ✓ Problem Solving
 - ✓ Technical Support
 - ✓ Communication
 - ✓ Teamwork
 - ✓ Adaptability
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LANGUAGES

English, IsiNdebele, Afrikaans

REFERENCES

Available upon request.